

ECON4200 - Hong Kong Statutory Minimum Wage Analysis Education

2026-04-24

setting up again

Load Libraries & Define Mappings

load lib

done load lib

load earning map

```
earnings_map <- tibble(  
  code = str_pad(1:40, width = 2, pad = "0"),  
  midpoint_hkd = c(1000,2500,3500,4500,5500,6500,7500,8500,9500,  
                   10500,11500,12500,13500,14500,15500,16500,17500,  
                   18500,19500,21000,23000,25000,27000,29000,31000,  
                   33000,35000,37000,39000,41000,43000,45000,47000,  
                   49000,55000,65000,75000,85000,95000,120000)  
)  
  
other_earnings_map <- tibble(  
  code = str_pad(1:11, width = 2, pad = "0"),  
  midpoint_hkd = c(500,1500,2500,3500,4500,5500,6500,7500,8500,9500,12000)  
)
```

done load earning map

Data Loading & Processing

Load all files + calculate est hourly wage same method used

done

ranges mapping

done

CPI Processing (Real Wages)

```
## [1] "2011 Q2 Reference CPI Value is: 78"
```

done

Add Real Wage to Main Dataset

done adding real wage

Testing to reconfirm is same as previous rmd file

```
## === Basic Data Overview ===

## Total rows in raw data: 1300115

## Total rows after processing: 1300115

## Rows in focus sample: 572003

## Years covered:

## # A tibble: 112 x 3
##   year quarter     n
##   <dbl>   <dbl> <int>
## 1  1985         1  4502
## 2  1985         2  4807
## 3  1985         3  4698
## 4  1985         4  4769
## 5  1986         1  4732
## 6  1986         2  4832
## 7  1986         3  4659
## 8  1986         4  4683
## 9  1987         1  4627
## 10 1987         2  4743
## # i 102 more rows

##

## === Key Variable Summary ===

##   hourly_wage    real_hourly_wage    weight    hours_main_employment
##   Min.      : 2.325   Min.      : 2.219   Min.      : 0.0003   Min.      : 8.00
##   1st Qu.: 27.170   1st Qu.: 37.459   1st Qu.: 0.0005   1st Qu.:40.00
##   Median : 45.208   Median : 54.929   Median : 0.0005   Median :48.00
##   Mean   : 69.905   Mean   : 83.719   Mean   :142.7027   Mean   :46.78
##   3rd Qu.: 79.114   3rd Qu.: 91.772   3rd Qu.: 0.0008   3rd Qu.:54.00
##   Max.    :3452.244   Max.    :4090.254   Max.    :775.6073   Max.    :99.00
## earnings_main_hkd
##   Min.      : 1000
##   1st Qu.: 5500
##   Median : 9500
##   Mean   : 13038
##   3rd Qu.: 15500
##   Max.    :120000

##

## === Missing Values (%) ===

## # A tibble: 19 x 2
##   Variable      `Missing_%`
##   <chr>         <dbl>
## 1 industry_underemployed    100
## 2 previous_industry         100
## 3 previous_occupation       100
## 4 unemployment_duration     100
## 5 leave_reason              100
## 6 intend_same_industry      100
## 7 intend_industry           100
## 8 intend_same_occupation    100
```

```
## 9 intend_occupation      100
## 10 hours_secondary      99.7
## 11 year_completed_study  94.5
## 12 parent_serial        73.3
## 13 length_employment    58.8
## 14 spouse_serial        56.6
## 15 work_place           53.8
## 16 relationship         24.0
## 17 foreign_helper       24.0
## 18 other_employment_7days 24.0
## 19 earnings_underemployed 1.84

##
## === Hourly Wage Distribution (Nominal) ===
```

Min	P5	Median	Mean	P95	Max	N
2.32	11.99	45.21	69.91	206.54	3452.24	572003

1. Check Earnings Mapping

```
## Unique earnings_main codes and their mapped values:
```

```
## # A tibble: 40 x 3
##   earnings_main earnings_main_hkd     n
##   <chr>          <dbl> <int>
## 1 01              1000 21222
## 2 02              2500 29452
## 3 03              3500 33208
## 4 04              4500 32707
## 5 05              5500 35875
## 6 06              6500 42288
## 7 07              7500 41219
## 8 08              8500 46906
## 9 09              9500 33480
## 10 10            10500 40181
## 11 11            11500 18784
## 12 12            12500 25518
## 13 13            13500 15513
## 14 14            14500  8794
## 15 15            15500 26255
## 16 16            16500  7180
## 17 17            17500  5605
## 18 18            18500  8614
## 19 19            19500  3158
## 20 20            21000 25134
## 21 21            23000  7187
## 22 22            25000 10608
## 23 23            27000  3379
## 24 24            29000  3022
## 25 25            31000 12607
## 26 26            33000  2149
## 27 27            35000  4402
## 28 28            37000  1147
## 29 29            39000  1011
```

```
## 30 30                                41000  5255
## # i 10 more rows
```

2. Check Weighting Factor

```
## Total weighted population (should be around Hong Kong's employed population):
```

```
## [1] 81626372
```

3. Sample of Raw vs Processed Data

yea this one isnt going to be same but uh it exist

year	quarter	earnings_main	earnings_main_hk	hours_main	employment_main	hourly_wage	real_hourly_wage
1988	3	02	2500	20	28.77	63.93	
2001	3	09	9500	35	62.47	69.74	
2010	1	06	6500	44	34.00	35.90	
1996	3	05	5500	44	28.77	31.68	
1998	3	07	7500	40	43.15	43.54	
2007	2	30	41000	39	241.95	276.31	
2000	3	12	12500	66	43.59	48.16	
1990	3	28	37000	45	189.23	346.21	
2004	4	22	25000	60	95.90	113.56	
1992	4	08	8500	66	29.64	43.68	

Now actual content

Cont. The aggregate data shows a continued improvement in wages and a reduction in the low-wage share around 2011, the effect appears modest in the overall population. This raises the question of whether the minimum wage had a stronger impact on specific vulnerable groups. Therefore, the next step is to conduct a more disaggregated analysis by examining subgroups such as: - Workers with lower education levels

Education Grouping (Handle code change in 2012)

From Q1 1985 to Q4 2011 Code Description 1 No schooling / pre-primary 2 Primary 3 Lower secondary 4 Upper secondary 5 Six form 6 Technical / vocational training 7 Tertiary (non-degree courses) 8 Tertiary (First degree courses) 9 Tertiary (Post-degree courses)

From Q1 2012 to current Code Description 1 No schooling / pre-primary 2 Primary 3 Lower secondary 4 Upper secondary 5 Post-secondary – diploma / certificate including craft 6 Post-secondary – sub-degree 7 Post-secondary – first degree 8 Post-secondary – post-degree

Then we have to do some recode mapping from numbers to words

Codes 1–4: No recoding needed, meanings identical across both periods

Code 5 (pre-2012): Kept as “Sixth Form”, this is academic pre-university, distinct from upper secondary (code 4)

Code 5 (post-2012): Labeled “Post-secondary Diploma/Certificate”, different meaning from pre-2012 code 5, so kept separate

Code 6 (pre-2012): Labeled “Technical / Vocational”, preserved as distinct category

Code 6 (post-2012): Labeled “Post-secondary Sub-degree”, different meaning from pre-2012 code 6, kept separate

Code 7 (pre-2012): Labeled “Tertiary Non-degree”, below bachelor level, preserved as distinct category

Code 7 (post-2012): Mapped to “Bachelor’s Degree”, conceptually equivalent to pre-2012 code 8

Code 8 (pre-2012): Mapped to “Bachelor’s Degree”, conceptually equivalent to post-2012 code 7

Code 8 (post-2012): Mapped to “Postgraduate”, conceptually equivalent to pre-2012 code 9

Code 9 (pre-2012): Mapped to “Postgraduate”, no equivalent in post-2012 scheme

Codes 99, 999, blank, NA: Coded as “Unknown / Missing”

Education distribution

Let’s check the education distribution:

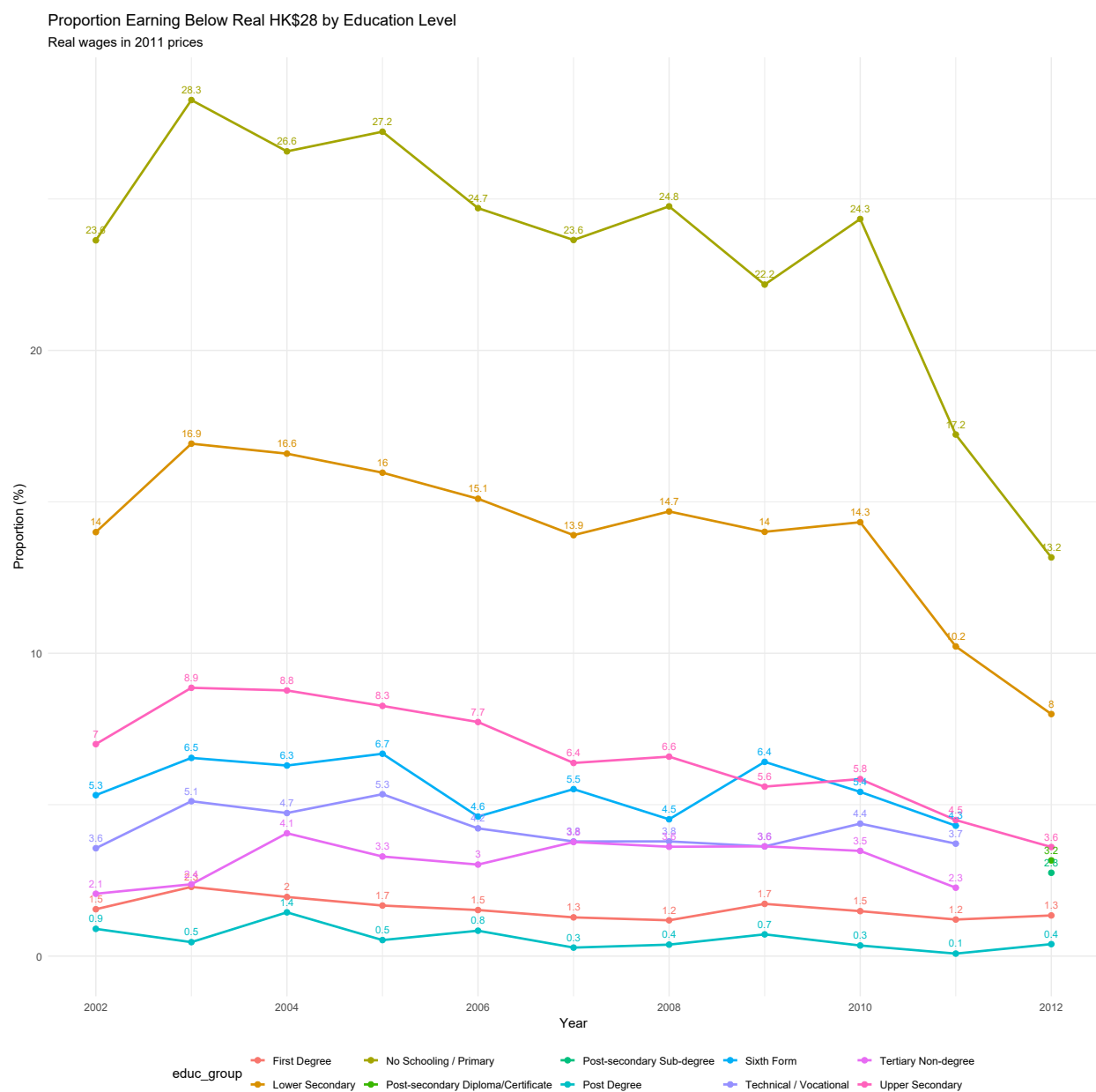
```
## # A tibble: 10 x 4
##   educ_group          educ_broad      n    pct
##   <chr>              <chr>      <int> <dbl>
## 1 Upper Secondary    Secondary + Vocational 194283 34.0
## 2 No Schooling / Primary Lower Education    113655 19.9
## 3 Lower Secondary    Lower Education    100911 17.6
## 4 First Degree        Tertiary (Degree)    69120 12.1
## 5 Tertiary Non-degree Secondary + Vocational  30148  5.27
## 6 Sixth Form          Secondary + Vocational  24911  4.36
## 7 Technical / Vocational Secondary + Vocational  21991  3.84
## 8 Post Degree          Tertiary (Degree)    14943  2.61
## 9 Post-secondary Diploma/Certificate Secondary + Vocational   1094  0.19
## 10 Post-secondary Sub-degree Secondary + Vocational    947  0.17
```

```
## Percentage of Unknown education in focus sample: 0 %
```

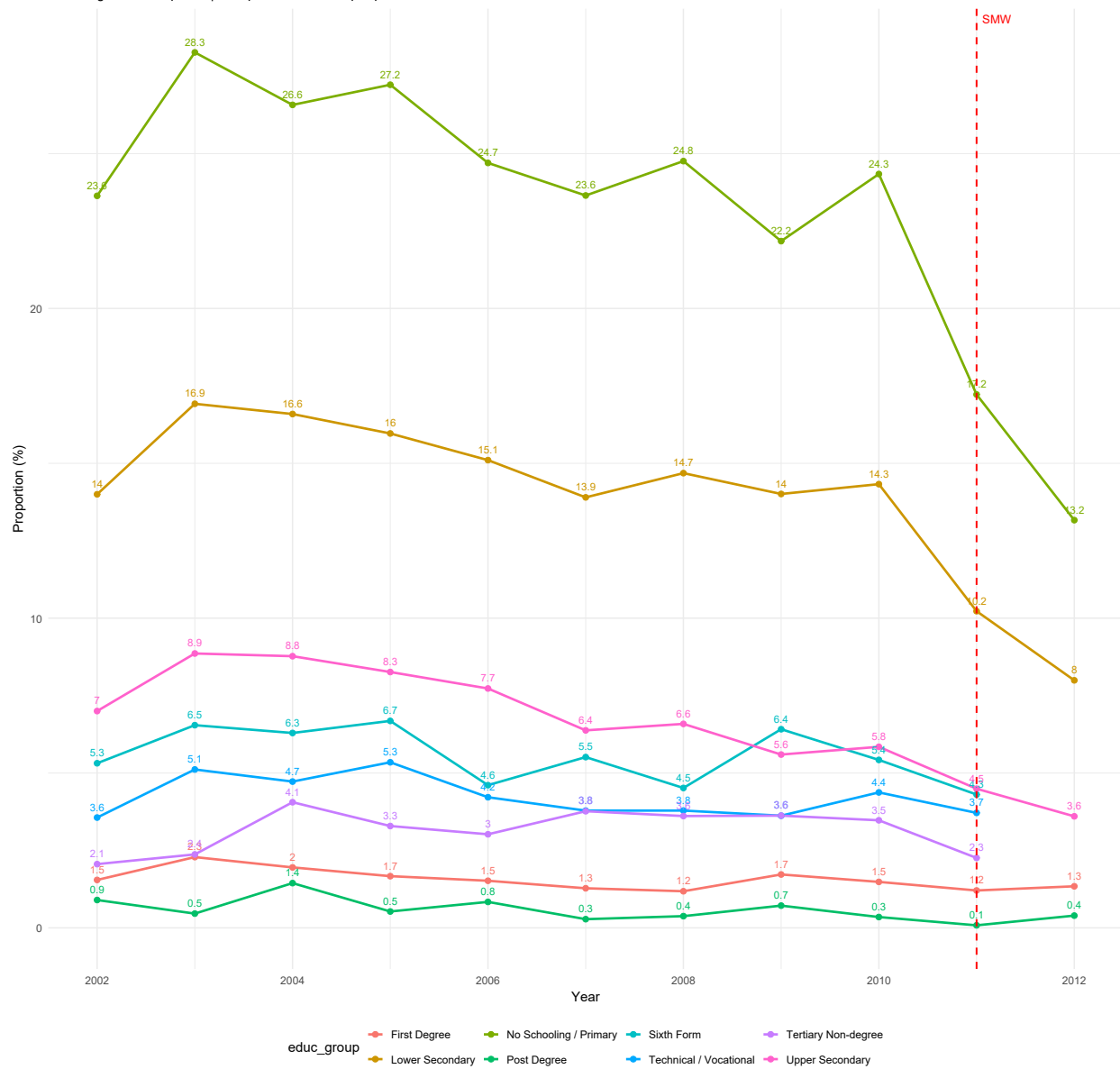
2.4% unknown edu, we can filter them to make final graphs/tables cleaner

```
## Percentage of Unknown education in focus sample after filtering yes this is going to be 0%: 0 %
```

Figure 1: Proportion Below HK 28 by Education Group



Proportion Earning Below Real HK\$28 by Education Level
Real wages in 2011 prices | Groups with sufficient pre/post data



Key Findings from the Education Analysis

Yes the changes are quite significant to the point you can directly use your eyes to see some big different but anyways

1. Strongest Impact on the Least Educated Workers

Workers with No Schooling / Primary education consistently had the highest share of low-wage earners in their own group, like maybe no schooling 100 people has like 25 people are wage < 28 hkd vs upper school maybe 100 people has like 5 people only are wage < 28 hkd etc

BUT the thing is, in 2011 compared to 2010 there is a big drop of proportion, like 25/100 ppl then 2011 become 17/100 which is quite some huge improvement

2. Limited Impact on Better-Educated Workers

I know i know the lines are “broke off” in 2012 but ignoring the broken line you can kinda know which is which

but either way that’s not the most important, the most important is that

Upper Secondary, Sixth Form, and Post-secondary groups had much lower exposure to wages below \$28 even before the policy.

Tertiary-educated workers (Bachelor’s, Postgraduate, etc.) had very low proportions ($< 3\text{--}4\%$) throughout the period, with almost no meaningful change after 2011.

3. The Policy Accelerated an Existing Trend

There was already a gradual downward trend in low-wage employment before 2011 (due to economic growth and structural changes) yes i know there are highs and ups and downs but the overall trend is decreasing.

However, the speed of decline increased noticeably in 2010 vs 2011, especially for No Schooling/Primary and Lower Secondary groups, consistent with the minimum wage taking effect in May 2011.

YoY % Change Tables and actual values by Education Group

Let’s see the YoY and actual of different edu group yes it is going to be $\% \text{ change} = (\text{new} - \text{old}) / \text{old} \times 100\%$
we see actual values cuz the graphs even i fig.width=10, fig.height=15, out.width=“100%” still has some data label overlapping each other

yes if they are some newly changed edu group in 2012 data then there is NOT going to be previous year comparison so they maybe NA on the % change, same reason if you are seeing removed and 2012 is gone

but we mainly care about the no schooling / primary AND lower secondary cuz 1. they are low edu 2. their % change are largest

```
##
## ## First Degree
##
##
##
## Table: Proportion Below Thresholds & YoY % Change - First Degree
##
## | year| Below $28| YoY $28| Below $35| YoY $35| Below $40| YoY $40|
## |----:|-----:|-----:|-----:|-----:|-----:|-----:|
## | 2008|      1.2|    -7.7|      2.2|    -8.3|      3.1|    -8.8|
## | 2009|      1.7|    41.7|      2.6|    18.2|      3.8|    22.6|
## | 2010|      1.5|   -11.8|      2.5|    -3.8|      3.5|    -7.9|
## | 2011|      1.2|   -20.0|      1.9|   -24.0|      2.8|   -20.0|
## | 2012|      1.3|     8.3|      2.3|    21.1|      2.9|     3.6|
##
##
## ## Lower Secondary
##
##
##
## Table: Proportion Below Thresholds & YoY % Change - Lower Secondary
##
## | year| Below $28| YoY $28| Below $35| YoY $35| Below $40| YoY $40|
## |----:|-----:|-----:|-----:|-----:|-----:|-----:|
```


##		2008	14.7	5.8	31.4	10.2	42.7	7.0
##		2009	14.0	-4.8	31.2	-0.6	43.4	1.6
##		2010	14.3	2.1	30.7	-1.6	42.6	-1.8
##		2011	10.2	-28.7	28.1	-8.5	40.8	-4.2
##		2012	8.0	-21.6	26.6	-5.3	39.5	-3.2

##

##

No Schooling / Primary

##

##

##

Table: Proportion Below Thresholds & YoY % Change - No Schooling / Primary

##

##		year	Below \$28	YoY \$28	Below \$35	YoY \$35	Below \$40	YoY \$40
##		----:	-----:	-----:	-----:	-----:	-----:	-----:
##		2008	24.8	5.1	44.4	8.6	54.1	4.4
##		2009	22.2	-10.5	43.9	-1.1	54.5	0.7
##		2010	24.3	9.5	42.4	-3.4	53.1	-2.6
##		2011	17.2	-29.2	39.0	-8.0	52.0	-2.1
##		2012	13.2	-23.3	37.2	-4.6	50.2	-3.5

##

##

Post-secondary Diploma/Certificate

##

##

##

Table: Proportion Below Thresholds & YoY % Change - Post-secondary Diploma/Certificate

##

##		year	Below \$28	YoY \$28	Below \$35	YoY \$35	Below \$40	YoY \$40
##		----:	-----:	-----:	-----:	-----:	-----:	-----:
##		2012	3.2	NA	8.4	NA	13.2	NA

##

##

Post-secondary Sub-degree

##

##

##

Table: Proportion Below Thresholds & YoY % Change - Post-secondary Sub-degree

##

##		year	Below \$28	YoY \$28	Below \$35	YoY \$35	Below \$40	YoY \$40
##		----:	-----:	-----:	-----:	-----:	-----:	-----:
##		2012	2.8	NA	7.1	NA	11.2	NA

##

##

Post Degree

##

##

##

Table: Proportion Below Thresholds & YoY % Change - Post Degree

##

##		year	Below \$28	YoY \$28	Below \$35	YoY \$35	Below \$40	YoY \$40
##		----:	-----:	-----:	-----:	-----:	-----:	-----:
##		2008	0.4	33.3	0.4	0.0	0.5	-37.5
##		2009	0.7	75.0	0.8	100.0	1.1	120.0

```
## | 2010|      0.3|   -57.1|      0.6|   -25.0|      1.0|    -9.1|
## | 2011|      0.1|   -66.7|      0.1|   -83.3|      0.2|   -80.0|
## | 2012|      0.4|   300.0|      0.8|   700.0|      0.9|   350.0|
```

```
##
##
```

```
## ## Sixth Form
```

```
##
##
##
```

```
## Table: Proportion Below Thresholds & YoY % Change - Sixth Form
```

```
##
```

```
## | year| Below $28| YoY $28| Below $35| YoY $35| Below $40| YoY $40|
## |----:|-----:|-----:|-----:|-----:|-----:|-----:|
## | 2008|      4.5|   -18.2|     11.4|    -0.9|     16.8|     2.4|
## | 2009|      6.4|    42.2|     13.2|    15.8|     18.7|    11.3|
## | 2010|      5.4|   -15.6|     10.8|   -18.2|     16.6|   -11.2|
## | 2011|      4.3|   -20.4|     10.1|    -6.5|     17.2|     3.6|
```

```
##
##
```

```
## ## Technical / Vocational
```

```
##
##
##
```

```
## Table: Proportion Below Thresholds & YoY % Change - Technical / Vocational
```

```
##
```

```
## | year| Below $28| YoY $28| Below $35| YoY $35| Below $40| YoY $40|
## |----:|-----:|-----:|-----:|-----:|-----:|-----:|
## | 2008|      3.8|     0.0|      8.5|     1.2|     14.1|     0.0|
## | 2009|      3.6|    -5.3|      8.5|     0.0|     14.2|     0.7|
## | 2010|      4.4|    22.2|     11.2|    31.8|     15.7|    10.6|
## | 2011|      3.7|   -15.9|      8.4|   -25.0|     15.2|    -3.2|
```

```
##
##
```

```
## ## Tertiary Non-degree
```

```
##
##
##
```

```
## Table: Proportion Below Thresholds & YoY % Change - Tertiary Non-degree
```

```
##
```

```
## | year| Below $28| YoY $28| Below $35| YoY $35| Below $40| YoY $40|
## |----:|-----:|-----:|-----:|-----:|-----:|-----:|
## | 2008|      3.6|    -5.3|      6.5|     8.3|      9.8|     7.7|
## | 2009|      3.6|     0.0|      6.1|    -6.2|      8.5|   -13.3|
## | 2010|      3.5|    -2.8|      6.0|    -1.6|     10.0|    17.6|
## | 2011|      2.3|   -34.3|      6.1|     1.7|     11.1|    11.0|
```

```
##
##
```

```
## ## Upper Secondary
```

```
##
##
##
```

```
## Table: Proportion Below Thresholds & YoY % Change - Upper Secondary
```

```
##
```

```
## | year| Below $28| YoY $28| Below $35| YoY $35| Below $40| YoY $40|
```

##	----	-----	-----	-----	-----	-----	-----
##	2008	6.6	3.1	15.8	6.0	24.5	6.5
##	2009	5.6	-15.2	15.0	-5.1	23.9	-2.4
##	2010	5.8	3.6	15.1	0.7	22.5	-5.9
##	2011	4.5	-22.4	13.0	-13.9	22.1	-1.8
##	2012	3.6	-20.0	11.9	-8.5	20.8	-5.9

Before vs After May 2011: Changes by Educational Attainment

To more directly assess the immediate impact of the Statutory Minimum Wage implemented in May 2011, we compare the proportion of workers earning below various hourly wage thresholds in the period **2008Q1–2011Q1 (Before)** VS **2011Q2–2012Q4 (After)**.

Why We Show 3 Thresholds (\$28, \$35, \$40) Instead of Just \$28

1. To show the depth of the policy’s impact

\$28 = The actual Statutory Minimum Wage (the legal floor).

\$35 and \$40 = Slightly higher thresholds to see if the policy also helped workers who were earning just above the minimum wage (a “ripple effect” or “spillover effect”).

If only the \$28 bar drops significantly while \$35 and \$40 stay almost the same, it means the minimum wage mainly pushed people up to the floor, but didn’t help much beyond that. If all three drop, it suggests a broader compression of the lower wage distribution.

2. To make the analysis more robust and convincing

Relying on only one threshold (\$28) is risky — it could be affected by measurement error or specific data quirks.

Showing multiple thresholds makes your conclusion stronger and more credible to readers/professors.

3. Academic / Policy Relevance

In minimum wage research, it’s very common to examine effects at:

The exact minimum wage 10–20% above the minimum wage

This helps answer: → “Did the policy only affect the very bottom, or did it lift a wider group of low-wage workers?”

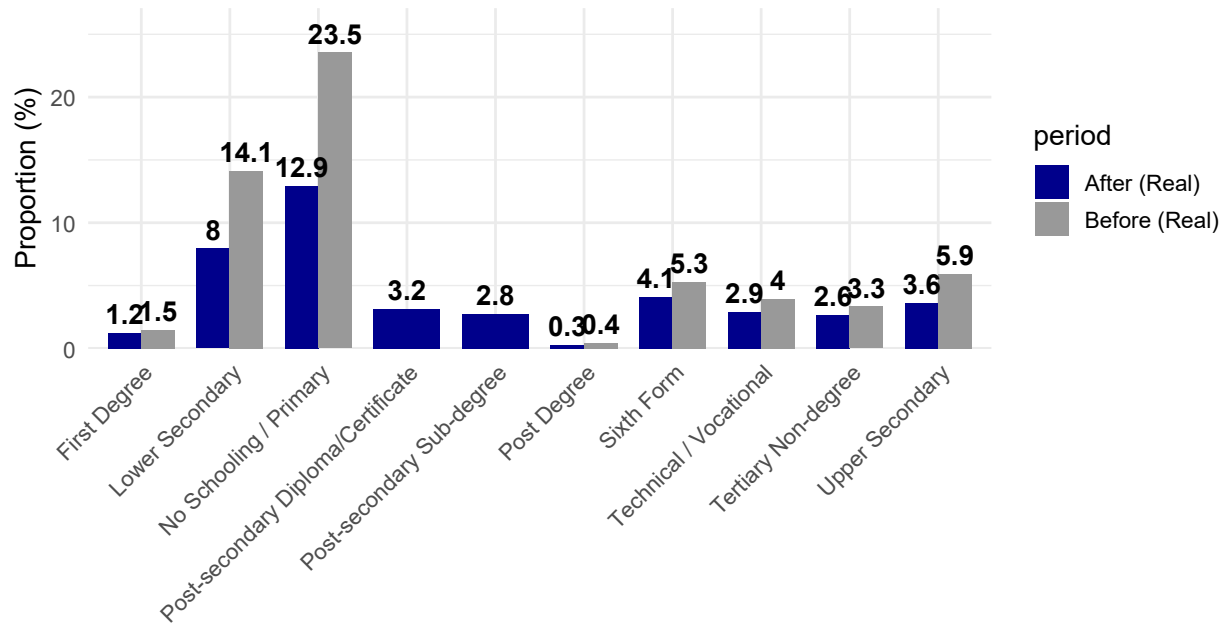
NO need to keep all three separate plots in the final report if space is limited.

Below \$28 graph

MAKE IT 2011Q1 VS 2011Q3

Below Real HK\$28

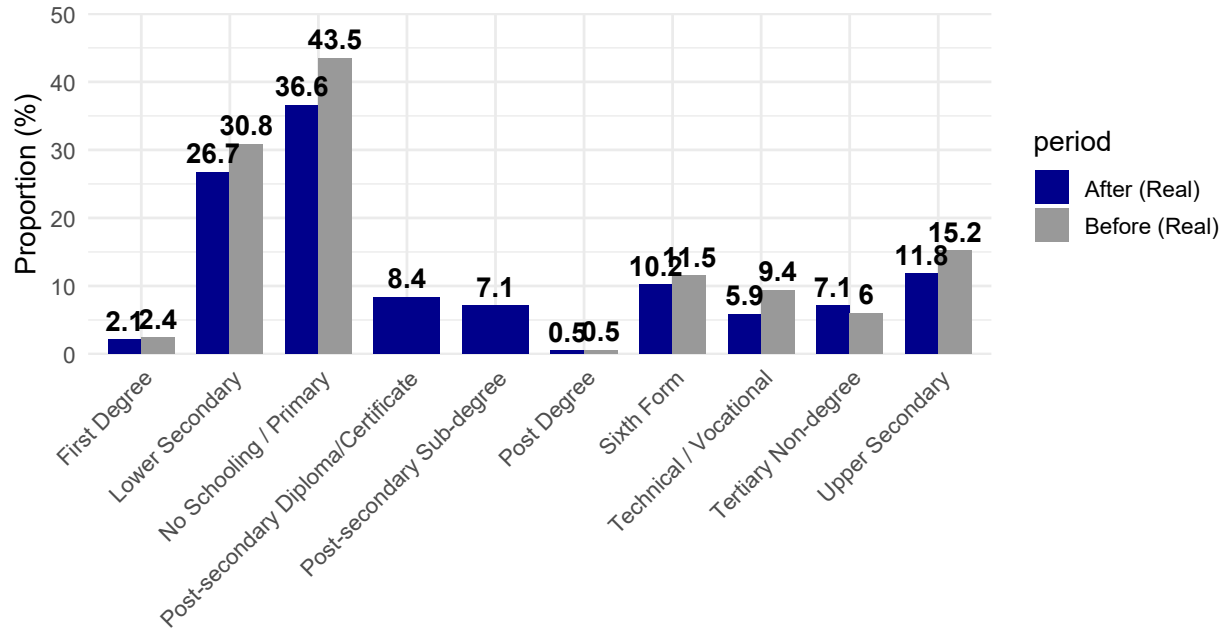
Before vs After May 2011



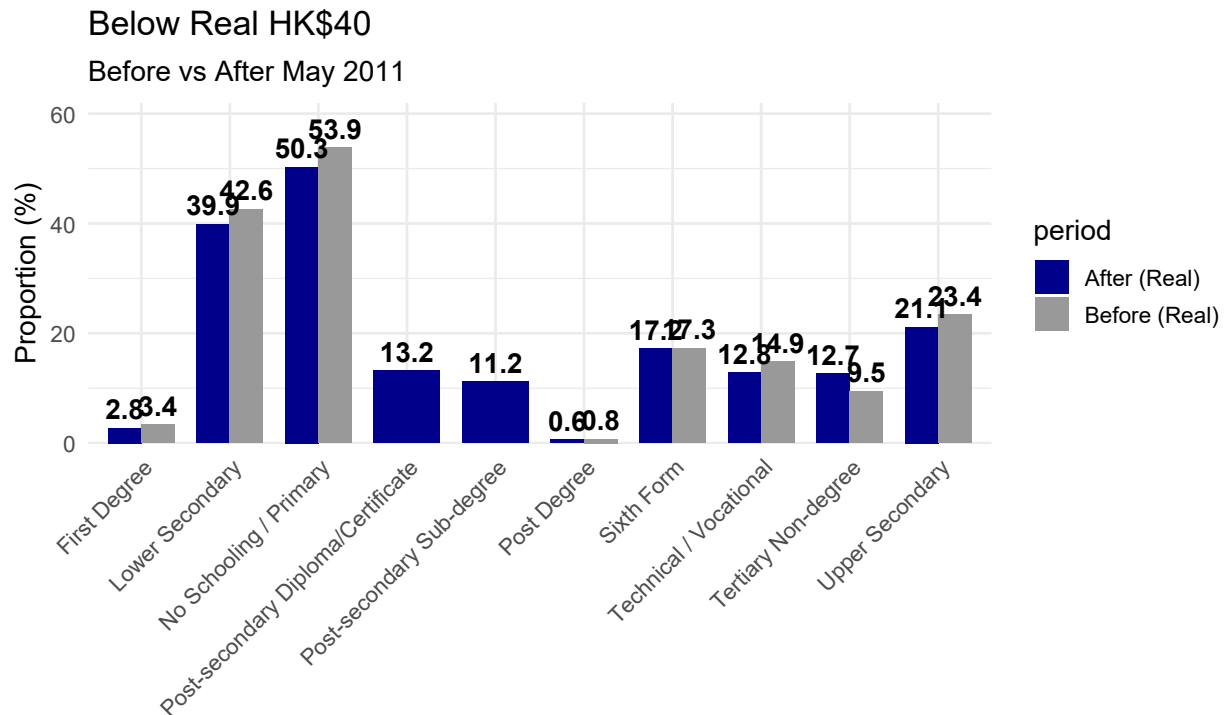
Below \$35 graph

Below Real HK\$35

Before vs After May 2011



Below \$40 graph



Summary Table: Before vs After by Education (Below \$28)

Yes it is EXTREMELY NORMAL THE RELATIVE CHANGE IS NOT EQUAL TO PREVIOUS YOY

Relative Change (%) Compares the average of the whole Before period (2008Q1–2011Q1) vs the whole After period (2011Q2–2012Q4). This is the overall policy impact.

YoY Tables shows year-to-year changes (e.g. 2010 → 2011, 2011 → 2012). These include normal economic fluctuations

Table 1: Before vs After May 2011: Proportion Below Real HK dollar 28 by Education Group

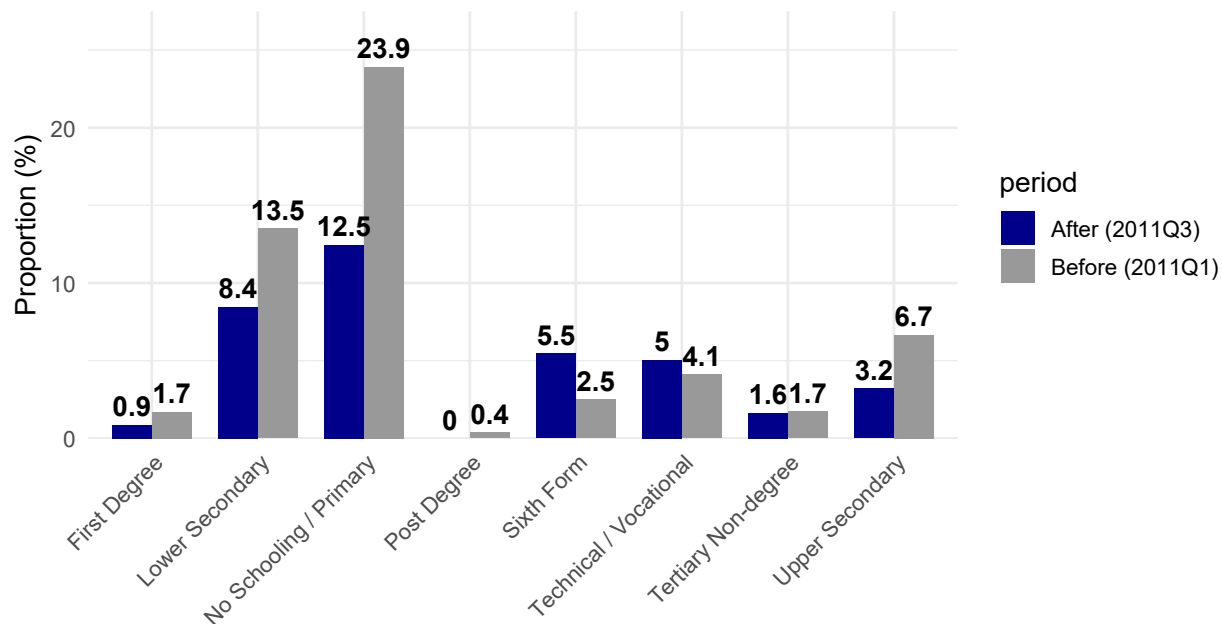
educ_group	After	Before	Change (pp)	Relative Change (%)
No Schooling / Primary	12.9	23.5	-10.6	-45.1
Lower Secondary	8.0	14.1	-6.1	-43.3
Upper Secondary	3.6	5.9	-2.3	-39.0
Sixth Form	4.1	5.3	-1.2	-22.6
Technical / Vocational	2.9	4.0	-1.1	-27.5
Tertiary Non-degree	2.6	3.3	-0.7	-21.2
First Degree	1.2	1.5	-0.3	-20.0
Post Degree	0.3	0.4	-0.1	-25.0
Post-secondary Diploma/Certificate	3.2	NA	NA	NA
Post-secondary Sub-degree	2.8	NA	NA	NA

The -45.1% in the summary table is more important for the conclusion because it shows the total effect of the policy.

RIGHT Before and After, 2011Q1 vs 2011Q3

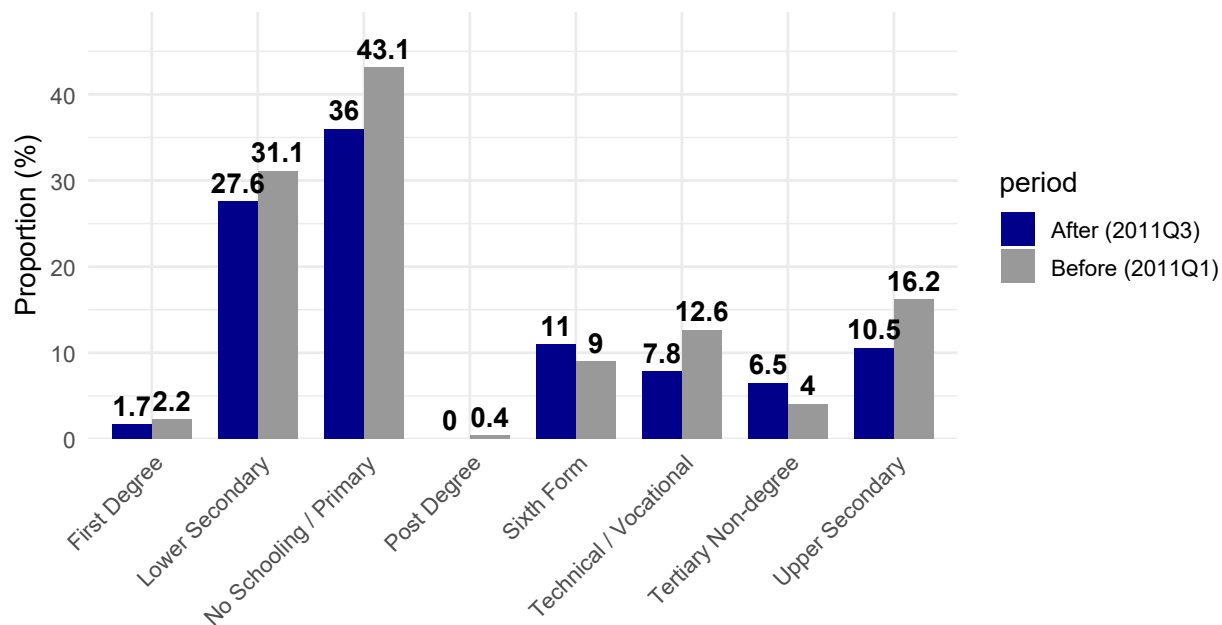
Below Real HK\$28

Right Before (2011Q1) vs Right After (2011Q3)



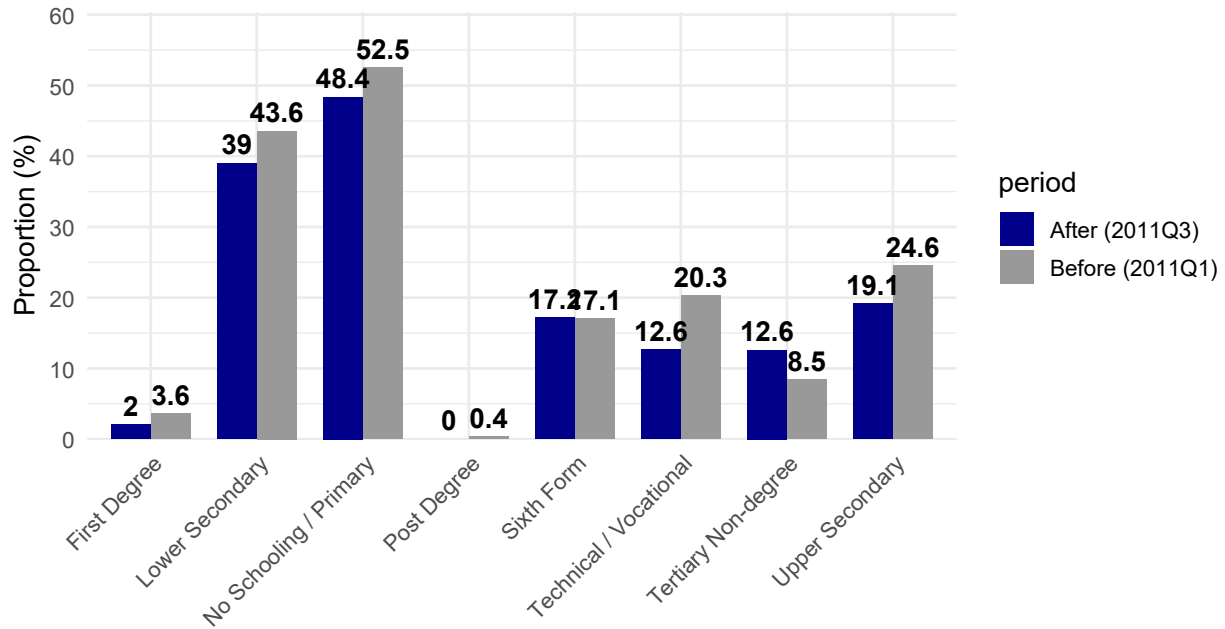
Below Real HK\$35

Right Before (2011Q1) vs Right After (2011Q3)



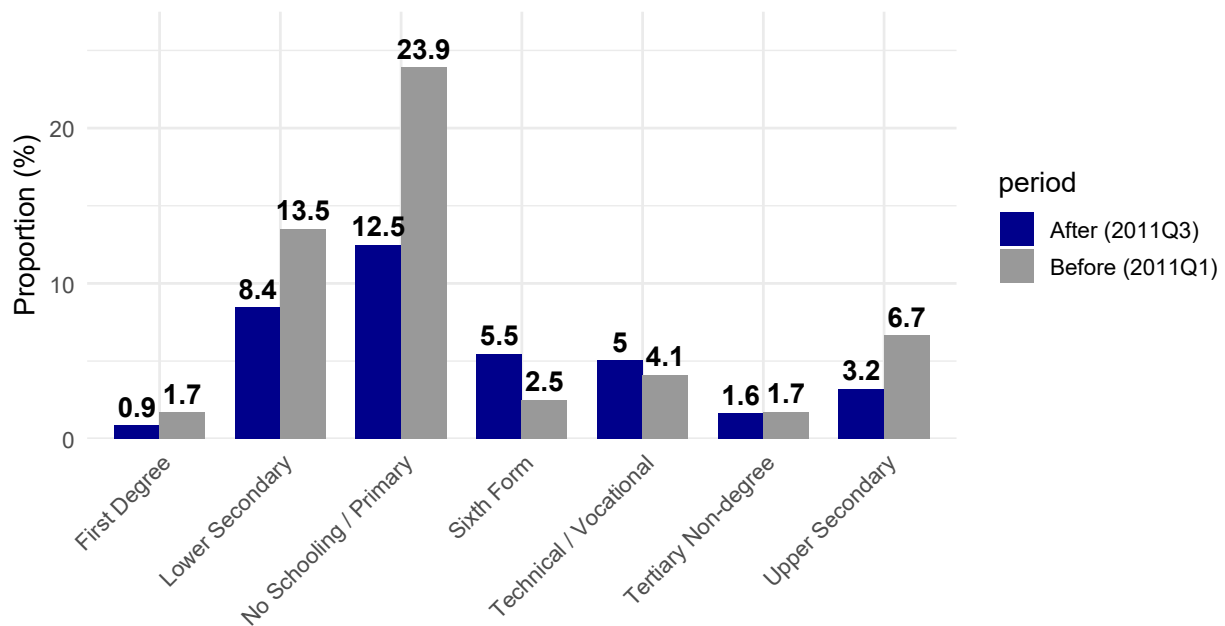
Below Real HK\$40

Right Before (2011Q1) vs Right After (2011Q3)



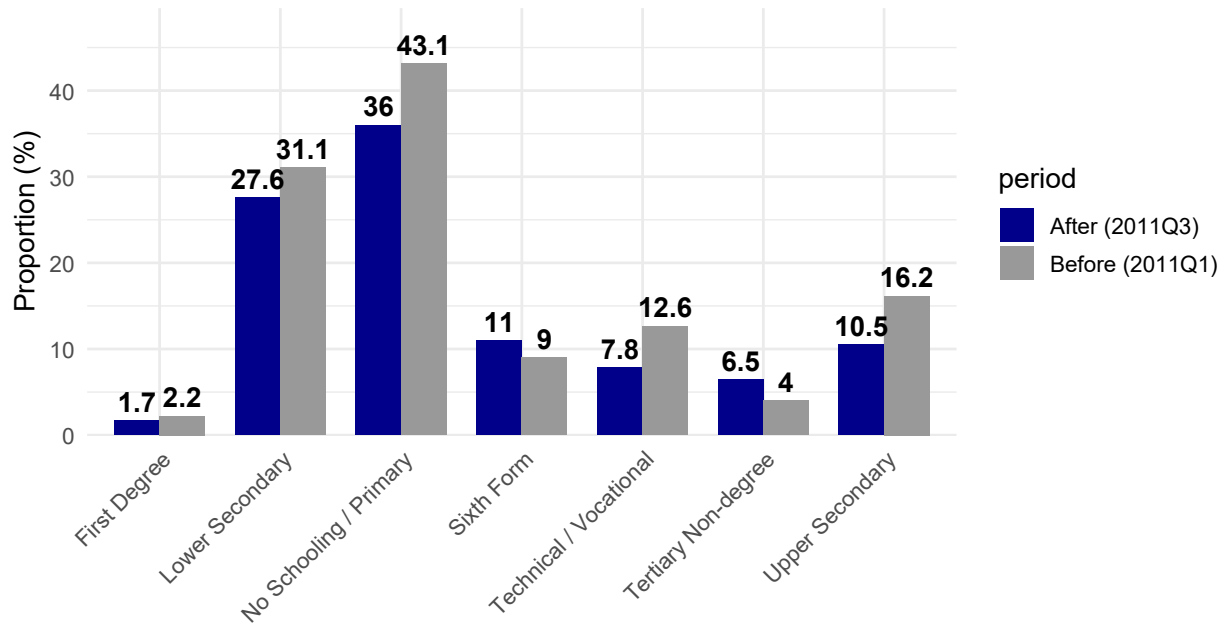
Below Real HK\$28 (Excluding Post Degree)

Right Before (2011Q1) vs Right After (2011Q3)



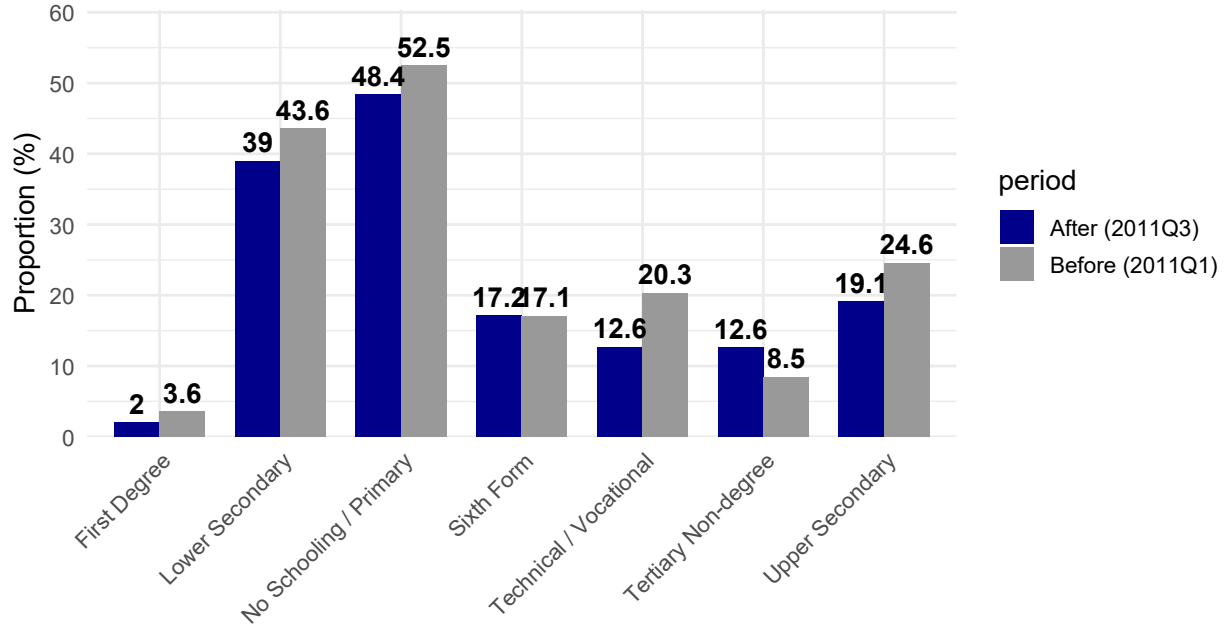
Below Real HK\$35 (Excluding Post Degree)

Right Before (2011Q1) vs Right After (2011Q3)



Below Real HK\$40 (Excluding Post Degree)

Right Before (2011Q1) vs Right After (2011Q3)



Summary table

Table 2: Right Before vs Right After (2011Q1 vs 2011Q3): Proportion Below HK dollar 28 by Education

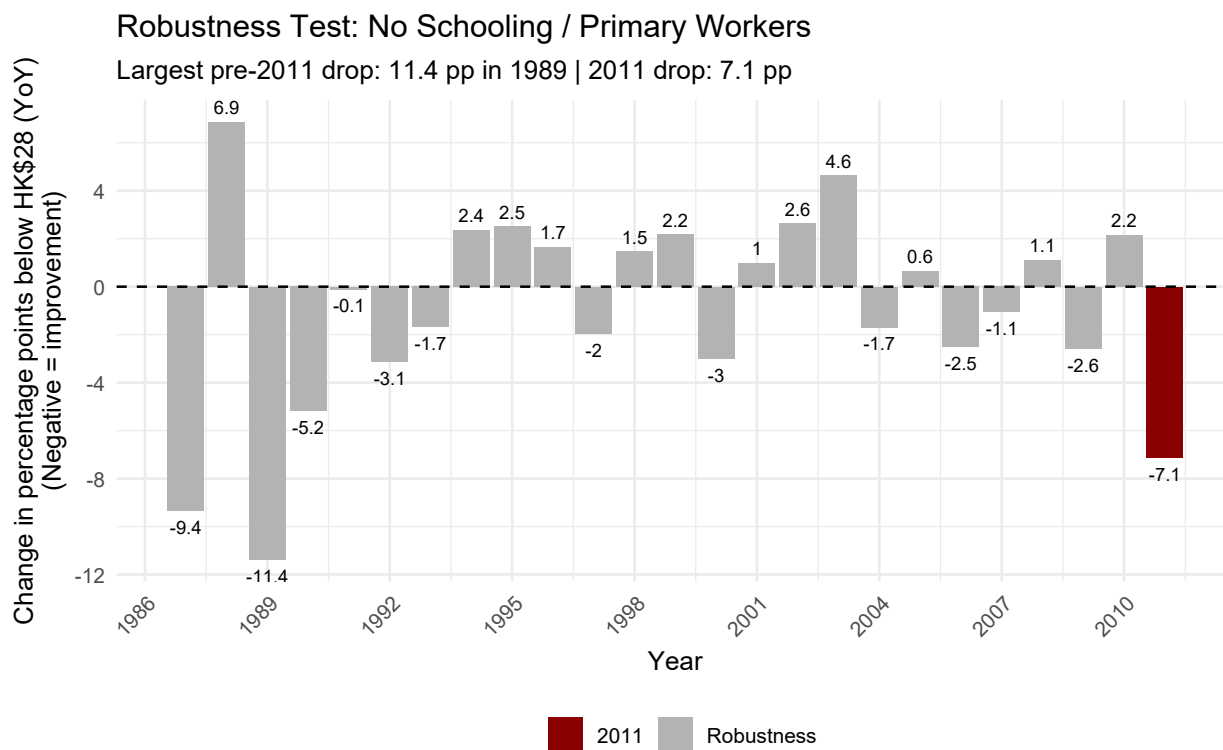
educ_group	After (2011Q3)	Before (2011Q1)	Change (pp)	Relative Change (%)
No Schooling / Primary	12.5	23.9	-11.4	-47.9
Lower Secondary	8.4	13.5	-5.1	-37.6
Upper Secondary	3.2	6.7	-3.4	-51.7
First Degree	0.9	1.7	-0.8	-49.8
Post Degree	0.0	0.4	-0.4	-100.0
Tertiary Non-degree	1.6	1.7	-0.1	-5.3
Technical / Vocational	5.0	4.1	0.9	22.7
Sixth Form	5.5	2.5	3.0	118.3

Robustness Tests for Education

Test 1: False Timing Robustness (Lowest Education Group)

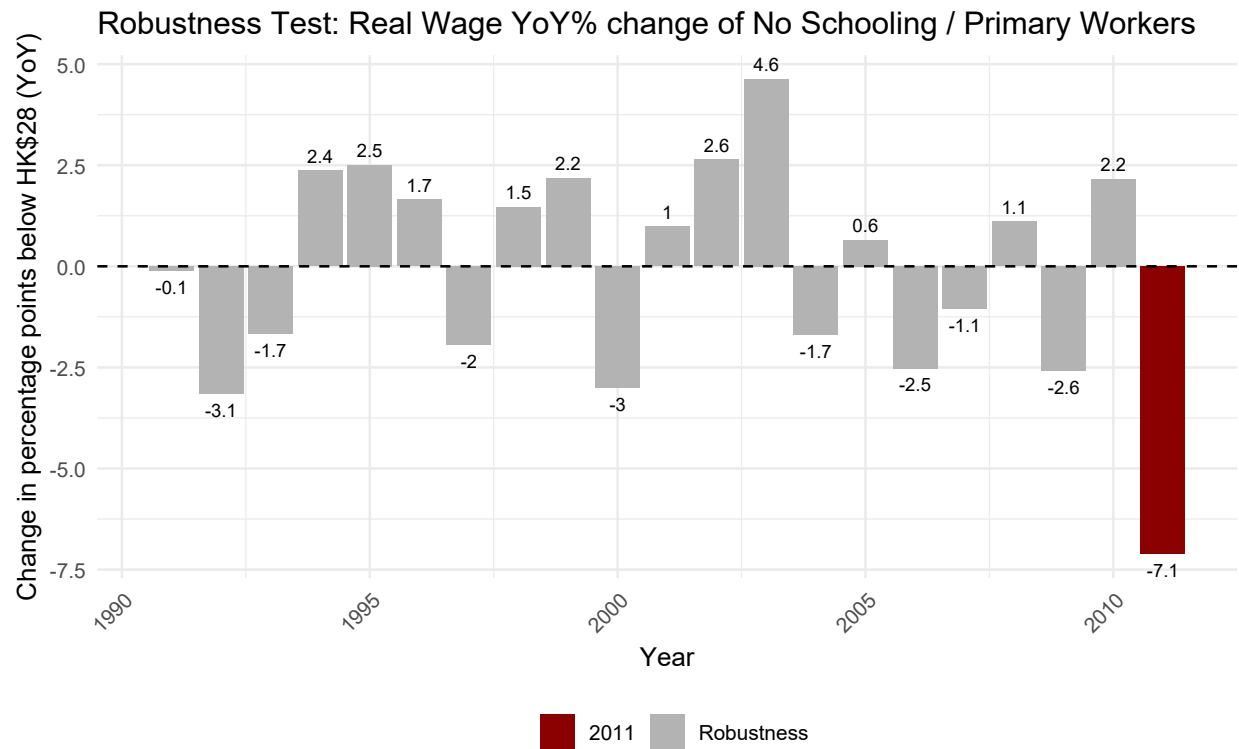
Test whether the 2011 drop for “No Schooling / Primary” workers was historically unusual:

```
##
## Largest pre-2011 drop:
## # A tibble: 1 x 2
##   largest_drop year_of_largest
##   <dbl>         <dbl>
## 1     -11.4         1989
```



```
##
## Largest pre-2011 drop:
```

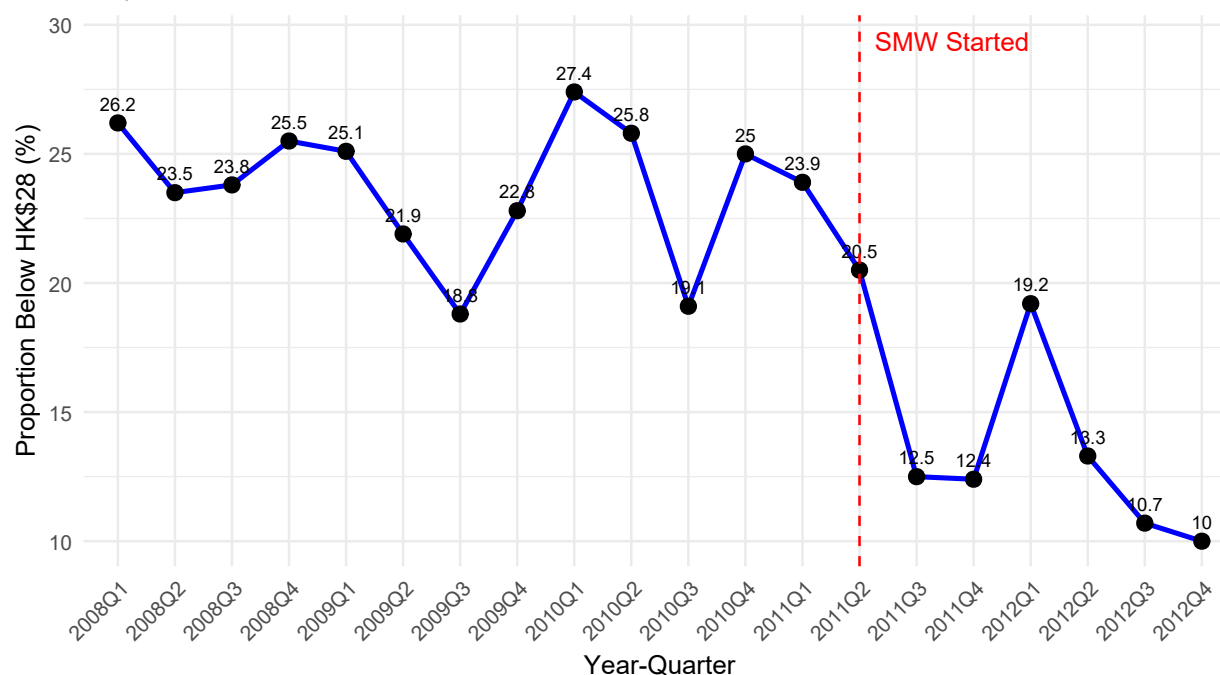
```
## # A tibble: 1 x 2
##   largest_drop year_of_largest
##   <dbl>         <dbl>
## 1      -3.15         1992
```



Test 2: Quarterly Robustness for Lowest Education Group

Quarterly Robustness Test: No Schooling / Primary Workers

Proportion Below HK\$28 Over Time



Small summary i guess?

1. Largest Impact on the Least Educated

Workers with No Schooling / Primary education experienced the most dramatic improvement. The proportion earning below HK\$28 dropped sharply from 23.5% (Before) to 12.9% (After), a reduction of over 10 percentage points.

Lower Secondary workers also saw a substantial decline (14.1% → 8.0%).

2. Lower Secondary workers also saw a substantial decline (14.1% → 8.0%).

For workers with Upper Secondary, Sixth Form, and Post-secondary qualifications, the reductions were much smaller.

Tertiary-educated workers (Bachelor's Degree and above) had very low exposure to sub-\$28 wages even before the policy, so the impact was minimal (as expected).

3. Clearer Effect at Lower Thresholds

The policy's effect is strongest at \$28, still visible at \$35, and weaker at \$40. This pattern confirms that the minimum wage primarily lifted the lowest-paid workers rather than causing a broad upward shift across all low-wage earners.

Summary?

The disaggregated analysis by educational attainment demonstrates that the Statutory Minimum Wage had a strongly progressive distributional impact.

The policy led to the largest reductions in the incidence of very low pay among workers with the lowest levels of education, aka those with No Schooling/Primary and Lower Secondary qualifications.

In the before-after comparison (2008Q1–2011Q1 vs 2011Q2–2012Q4), these groups experienced the most substantial declines in the proportion earning below HK\$28 (real terms).

In contrast, the effect on tertiary-educated workers was negligible, as expected given their already higher wage levels.

These findings suggest that the minimum wage was relatively successful in targeting and improving labour market outcomes for the most vulnerable segments of the Hong Kong workforce.

Limitations

While the before-after comparison and disaggregated analysis by education show a clear reduction in low-wage employment coinciding with the implementation of the minimum wage, several limitations should be noted.

First, the Labour Force Survey is a repeated cross-sectional dataset, meaning we cannot track the same individuals over time.

Second, the period 2011–2012 also coincided with economic recovery, which may have independently contributed to wage growth.

Third, because the minimum wage was announced in advance, some employers may have adjusted wages even before May 2011, potentially attenuating the observed before-after difference.

Finally, the use of midpoint approximations for earnings bands and setting the weeks of each month to be 4.345 introduces some measurement error in hourly wage estimates, yes remember:

(hours_main_employment or hours_7days) * 4.345 is monthly working hour

Then we are doing monthly MIDPOINT EARNINGS / monthly working hour to get the estimated hourly wage, then this

Sidenote

Statistical significance is irrelevant. The question is whether the magnitude of the change is economically meaningful — and a drop from 23.5% to 12.9% clearly is.

This is actually more rigorous than tacking on p-values. Many economists (especially in labor economics) argue that with large administrative or survey data, significance testing is misleading because everything is significant.

Robustness check (discarded)

1. Sample Composition Check (Addresses “Did low-skill workers disappear?”)

```
## # A tibble: 10 x 4
##   year educ_group      weighted_n raw_n
##   <dbl> <chr>          <dbl> <int>
## 1 2008 Lower Secondary      2.00  3651
## 2 2008 No Schooling / Primary 1.46  2750
## 3 2009 Lower Secondary      1.95  3543
## 4 2009 No Schooling / Primary 1.36  2525
## 5 2010 Lower Secondary      1.90  3400
## 6 2010 No Schooling / Primary 1.26  2321
```

```
## 7 2011 Lower Secondary      1.89 3350
## 8 2011 No Schooling / Primary 1.30 2344
## 9 2012 Lower Secondary      1.91 3276
## 10 2012 No Schooling / Primary 1.28 2219
```

Weighted N for No Schooling/Primary: 1.46M (2008) → 1.30M (2011) = -11% decline

Weighted N for Lower Secondary: 2.00M → 1.89M = -5.5% decline

Some low-skill workers left the sample (possible)

But decline is modest (not massive)

Could explain part of the proportion drop (Possible)

But not all — drop was 10+ percentage points

What to look for: If weighted_n drops sharply in 2011-2012, composition bias is possible. If it's stable, you can argue against it.

2. Upper Bound Robustness Test (Addresses measurement error)

```
## # A tibble: 5 x 3
##   year below_28 below_80
##   <dbl>   <dbl>   <dbl>
## 1 2008    0.0834    0.606
## 2 2009    0.0763    0.595
## 3 2010    0.0770    0.606
## 4 2011    0.0557    0.600
## 5 2012    0.0448    0.596
```

Below \$28: 8.77% → 5.93% = -2.84pp drop

Below \$80: 63.5% → 63.4% = essentially flat (then drops to 59.6% in 2012)

3. High-Education Control Group (Addresses “was it just the economy?”)

```
## # A tibble: 10 x 3
##   year educ_broad      below_28
##   <dbl> <chr>         <dbl>
## 1 2008 Lower Education  18.9
## 2 2008 Tertiary (Degree) 1.00
## 3 2009 Lower Education  17.4
## 4 2009 Tertiary (Degree) 1.48
## 5 2010 Lower Education  18.3
## 6 2010 Tertiary (Degree) 1.25
## 7 2011 Lower Education  13.1
## 8 2011 Tertiary (Degree) 0.956
## 9 2012 Lower Education  10.1
## 10 2012 Tertiary (Degree) 1.13
```

Sample Composition Bias (Did low-skill workers disappear?)

```
## # A tibble: 16 x 5
## # Groups:   educ_group [2]
##   year educ_group      weighted_count sample_n index_2008
##   <dbl> <chr>         <dbl>     <int>     <dbl>
```

##	1	2005 Lower Secondary	1.95	3771	97.5
##	2	2005 No Schooling / Primary	1.47	2936	101.
##	3	2006 Lower Secondary	1.95	3584	97.6
##	4	2006 No Schooling / Primary	1.49	2842	103.
##	5	2007 Lower Secondary	1.97	3684	98.5
##	6	2007 No Schooling / Primary	1.45	2793	99.5
##	7	2008 Lower Secondary	2.00	3651	100
##	8	2008 No Schooling / Primary	1.46	2750	100
##	9	2009 Lower Secondary	1.95	3543	97.5
##	10	2009 No Schooling / Primary	1.36	2525	93.0
##	11	2010 Lower Secondary	1.90	3400	95.1
##	12	2010 No Schooling / Primary	1.26	2321	86.5
##	13	2011 Lower Secondary	1.89	3350	94.5
##	14	2011 No Schooling / Primary	1.30	2344	89.3
##	15	2012 Lower Secondary	1.91	3276	95.8
##	16	2012 No Schooling / Primary	1.28	2219	87.6

Key observations:

No Schooling/Primary dropped ~10-13% from 2008 -> Some composition change occurred

But most of the drop happened BEFORE 2011 (2008→2010: -13.5%) -> Not a 2011-specific phenomenon

2010→2011: 86.5 → 89.3 (actually increased slightly) -> No evidence of mass layoffs in 2011

Lower Secondary remained stable throughout (95-100) -> No composition bias for this group 2008
=100 2005-2012

no schooling/primary 2008 to 2010 -> -13.5%, 2008 to 2011 -> -10.7%, 2011 2010

lower secondary, 2008 to 2010 -> -4.9%, 2008 to 2011 -> -5.5%,

2008-2010

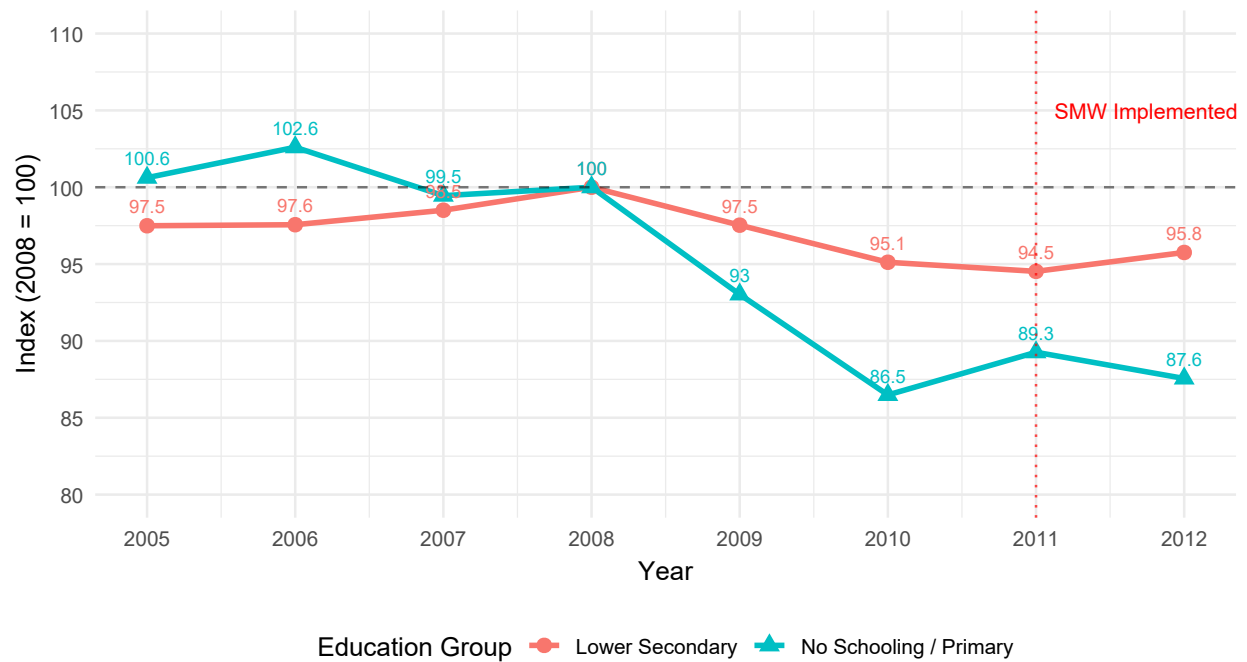
2010→2011

-10.6 -11%

‘ , ‘

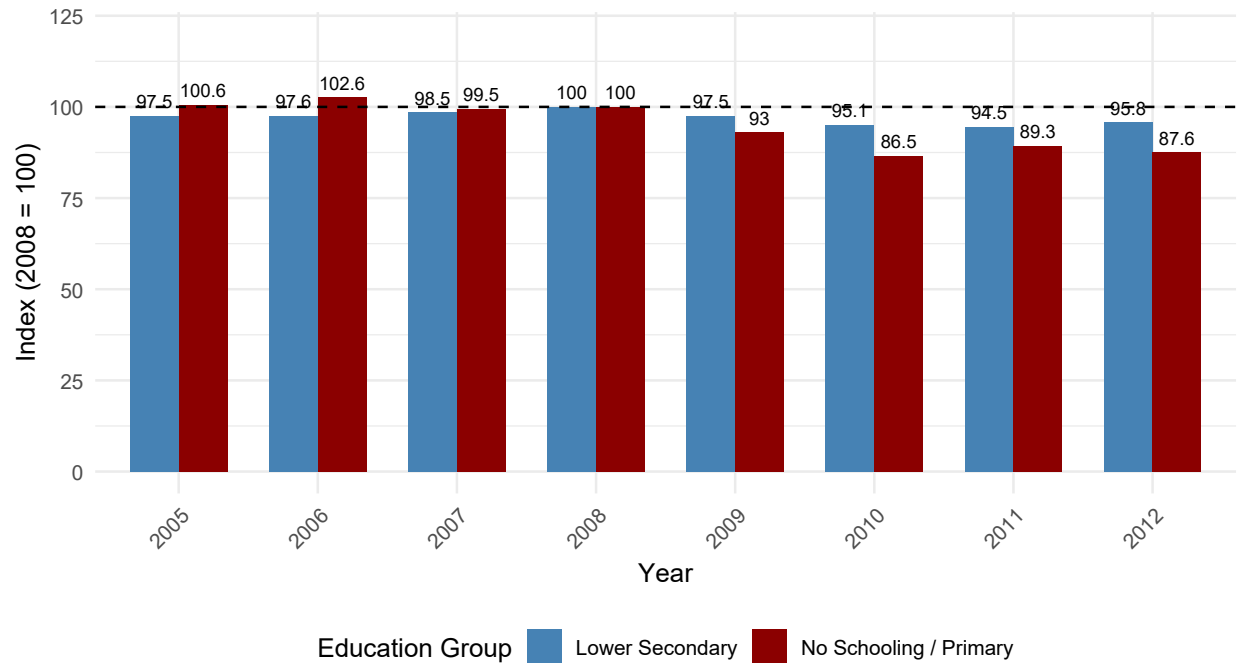
Sample Composition of Low-Education Groups (2008 = 100)

Weighted employment counts indexed to 2008 baseline



Sample Composition of Low-Education Groups (2008 = 100)

Weighted employment counts indexed to 2008 baseline



Midpoint Error (Upper/Lower Bound Test)

A tibble: 5 x 4

##	year	midpoint	lower_bound	upper_bound
##	<dbl>	<dbl>	<dbl>	<dbl>
## 1	2008	32.6	38.3	23.2
## 2	2009	30.1	36.2	21.2
## 3	2010	28.6	34.2	19.5
## 4	2011	17.9	24.0	10.6
## 5	2012	10.6	16.2	5.71

schooling/ HK\$28

Midpoint , Lower Bound), code 01 = HK\$0 Upper Bound , code 01 =

HK\$1,999

4.345

2010→2011

28.6% → 17.9% -10.7

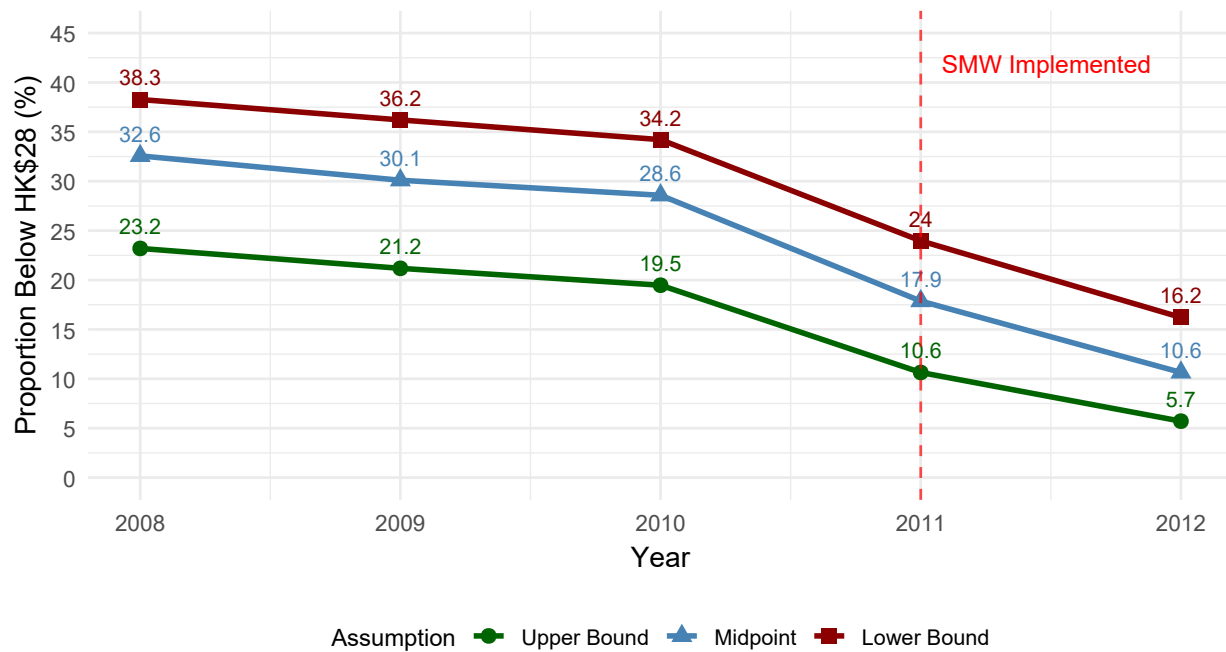
34.2% → 24.0% -10.2

19.5% → 10.6% -8.9

2011

Sensitivity Test: Proportion Below HK\$28 by Earnings Assumption

No Schooling / Primary Workers | Three methods of calculating hourly wage



All three methods show a large decline in 2011 => Results are NOT sensitive to midpoint choice

Midpoint: 28.6% → 17.9% (-10.7pp) => main result

Lower bound: 34.2% → 24.0% (-10.2pp) => Similar magnitude

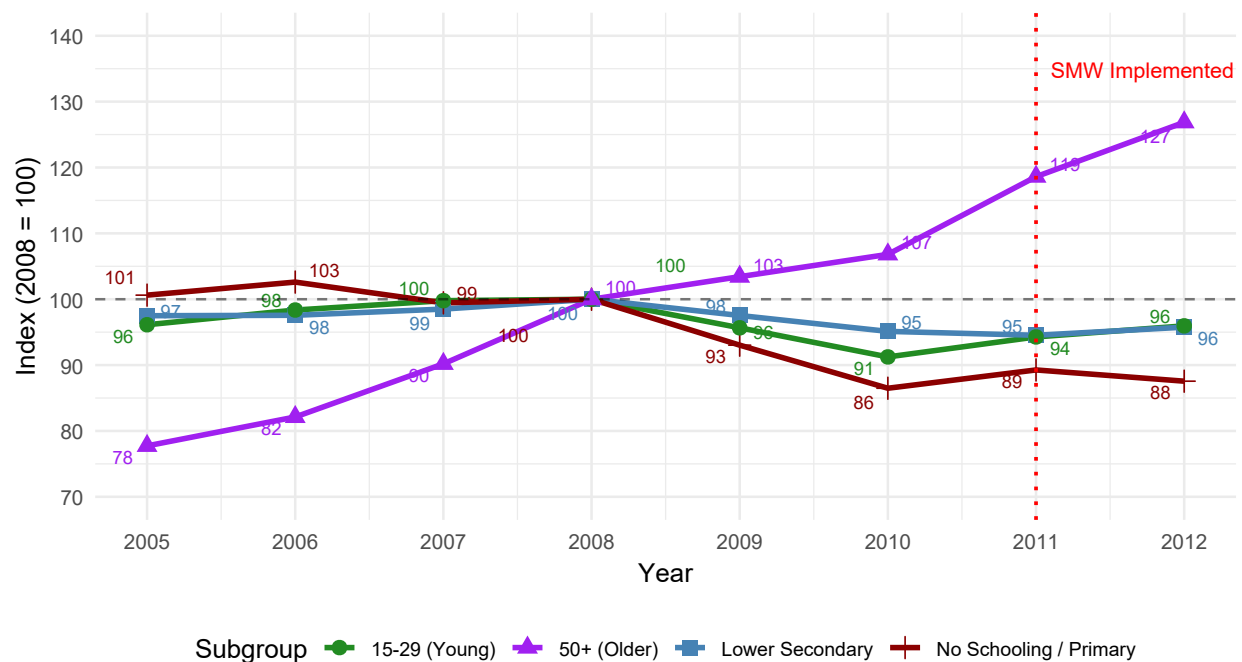
Upper bound: 19.5% → 10.6% (-8.9pp) => Slightly smaller but still large

The spread between bounds narrowed in 2011-2012 => Suggests less measurement error at lower wages

```
## # A tibble: 8 x 4
## # Groups:   subgroup [1]
##   year subgroup   index_2008 group_type
##   <dbl> <chr>         <dbl> <chr>
## 1  2005 50+ (Older)      77.7 Age Group
## 2  2006 50+ (Older)      82.1 Age Group
## 3  2007 50+ (Older)      90.2 Age Group
## 4  2008 50+ (Older)     100 Age Group
## 5  2009 50+ (Older)     103 Age Group
## 6  2010 50+ (Older)     107 Age Group
## 7  2011 50+ (Older)     119 Age Group
## 8  2012 50+ (Older)     127 Age Group
```

Sample Composition of Vulnerable Groups (2008 = 100)

Weighted employment counts indexed to 2008 baseline



No Schooling / Primary =

Lower Secondary =

15-29 (Young) = 2012

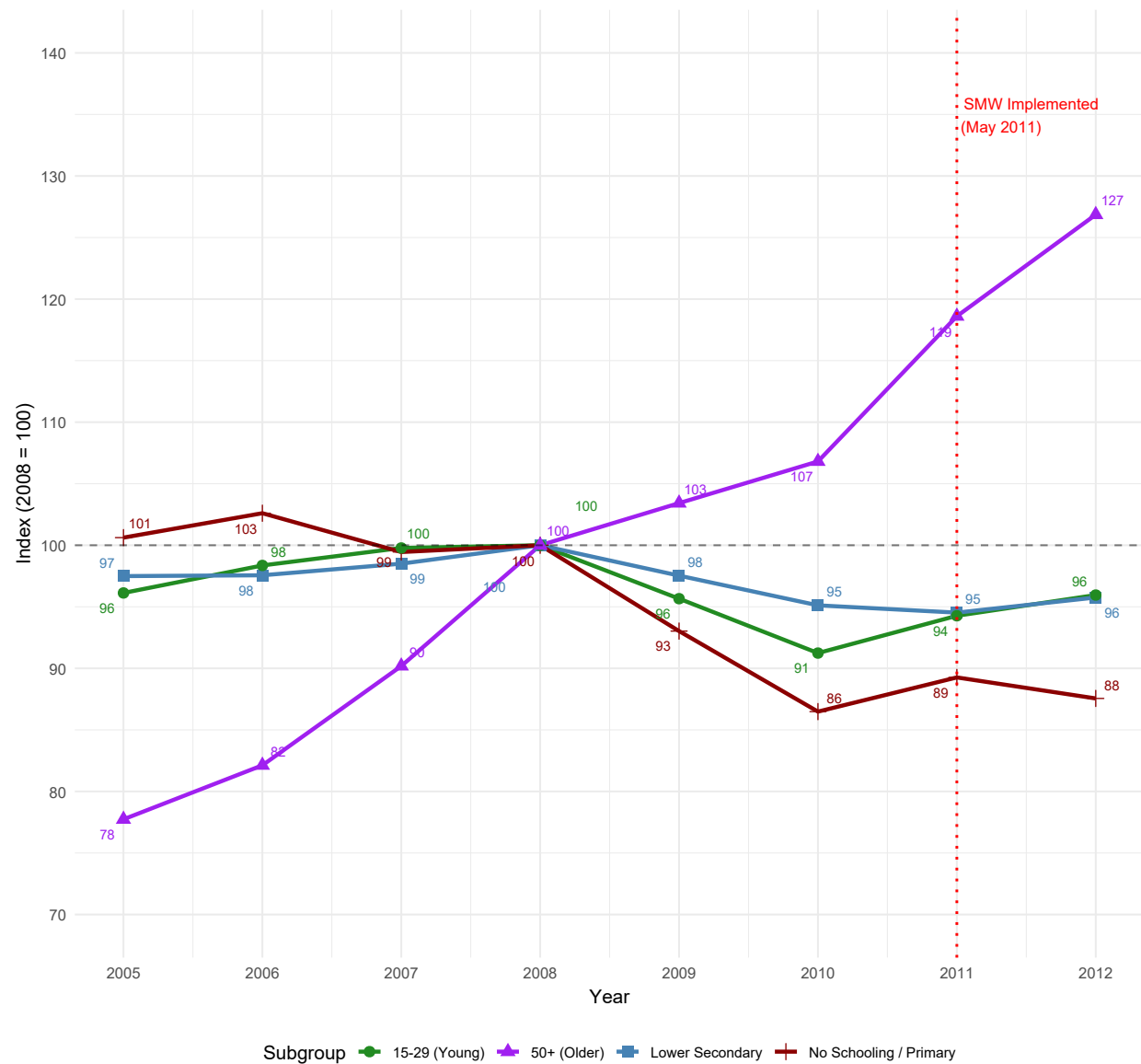
50+ (Older) =

10%
2008-2010 2011

17-30%
 ? ->
 2011 ? 2011 ->
 ? 30% -> /
 2011

Sample Composition of Vulnerable Groups (2008 = 100)

Weighted employment counts indexed to 2008 baseline

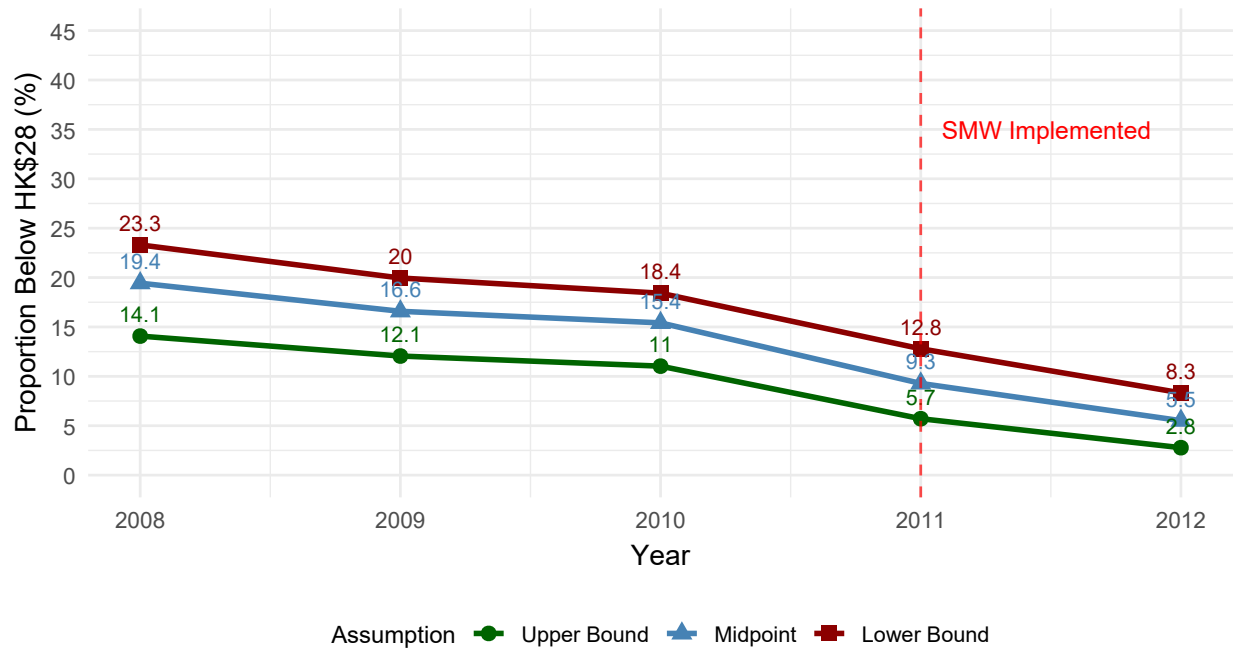


```
## # A tibble: 5 x 4
##   year midpoint lower_bound upper_bound
##   <dbl>      <dbl>      <dbl>      <dbl>
## 1  2008    19.4      23.3      14.1
## 2  2009    16.6      20.0      12.1
```

```
## 3 2010 15.4 18.4 11.0
## 4 2011 9.28 12.8 5.71
## 5 2012 5.51 8.32 2.78
```

Sensitivity Test: Proportion Below HK\$28 by Earnings Assumption

Older Workers (50+) | Three methods of calculating hourly wage



```
## # A tibble: 13 x 4
##   year below_23 below_28 below_33
##   <dbl>   <dbl>   <dbl>   <dbl>
## 1 2000 10.7    20.0    29.5
## 2 2001 9.55    21.0    31.1
## 3 2002 12.1    23.6    35.3
## 4 2003 14.4    28.3    38.8
## 5 2004 14.6    26.6    38.3
## 6 2005 14.9    27.2    40.2
## 7 2006 11.6    24.7    37.4
## 8 2007 10.8    23.6    38.0
## 9 2008 12.3    24.8    38.0
## 10 2009 9.77    22.2    36.1
## 11 2010 9.74    24.3    37.7
## 12 2011 6.04    17.2    35.0
## 13 2012 4.53    13.2    29.3
```

Proportion Below Different Thresholds (HK\$23, HK\$28, HK\$33)

No Schooling / Primary Workers | Real wages adjusted to 2011 Q2 prices

